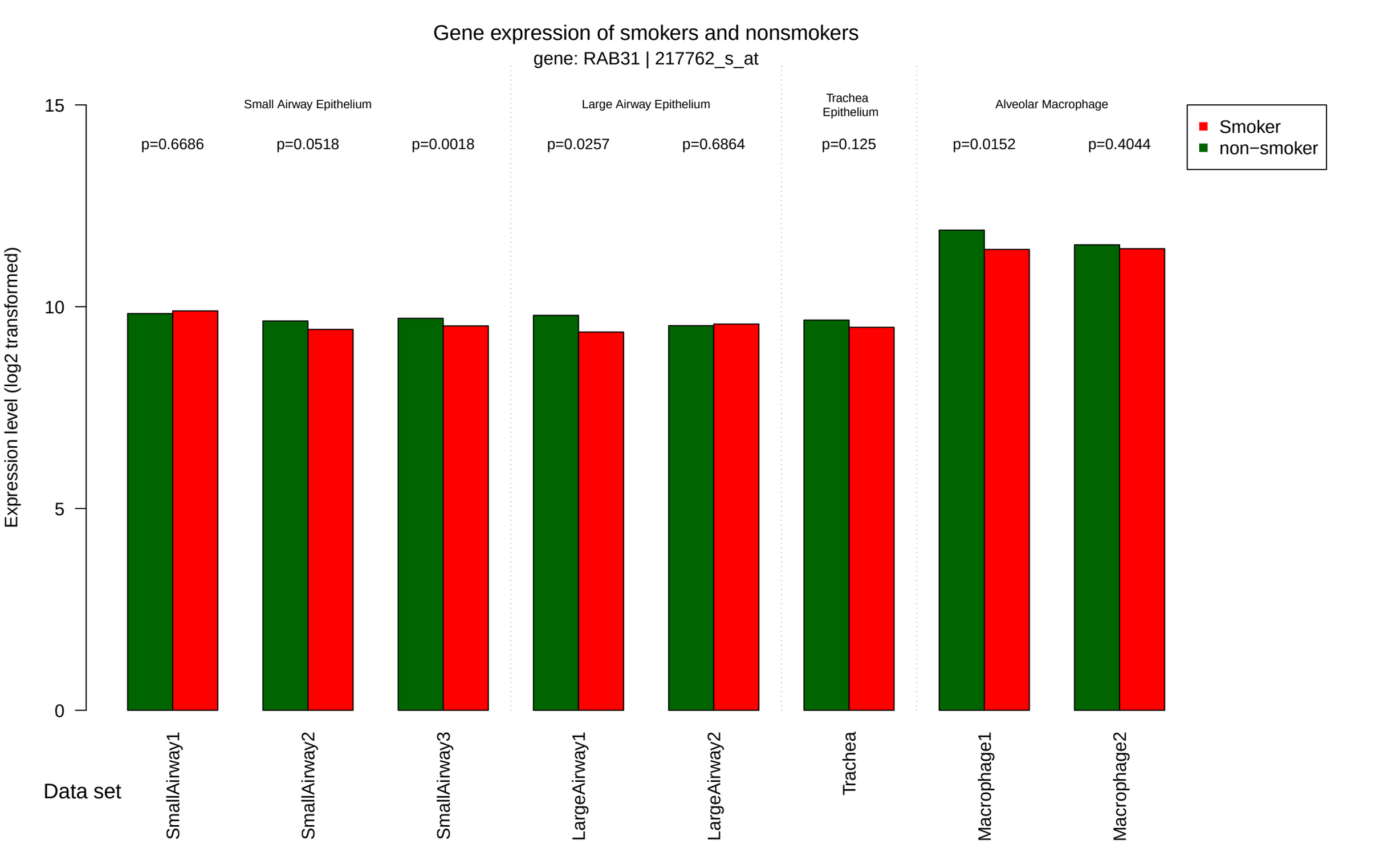
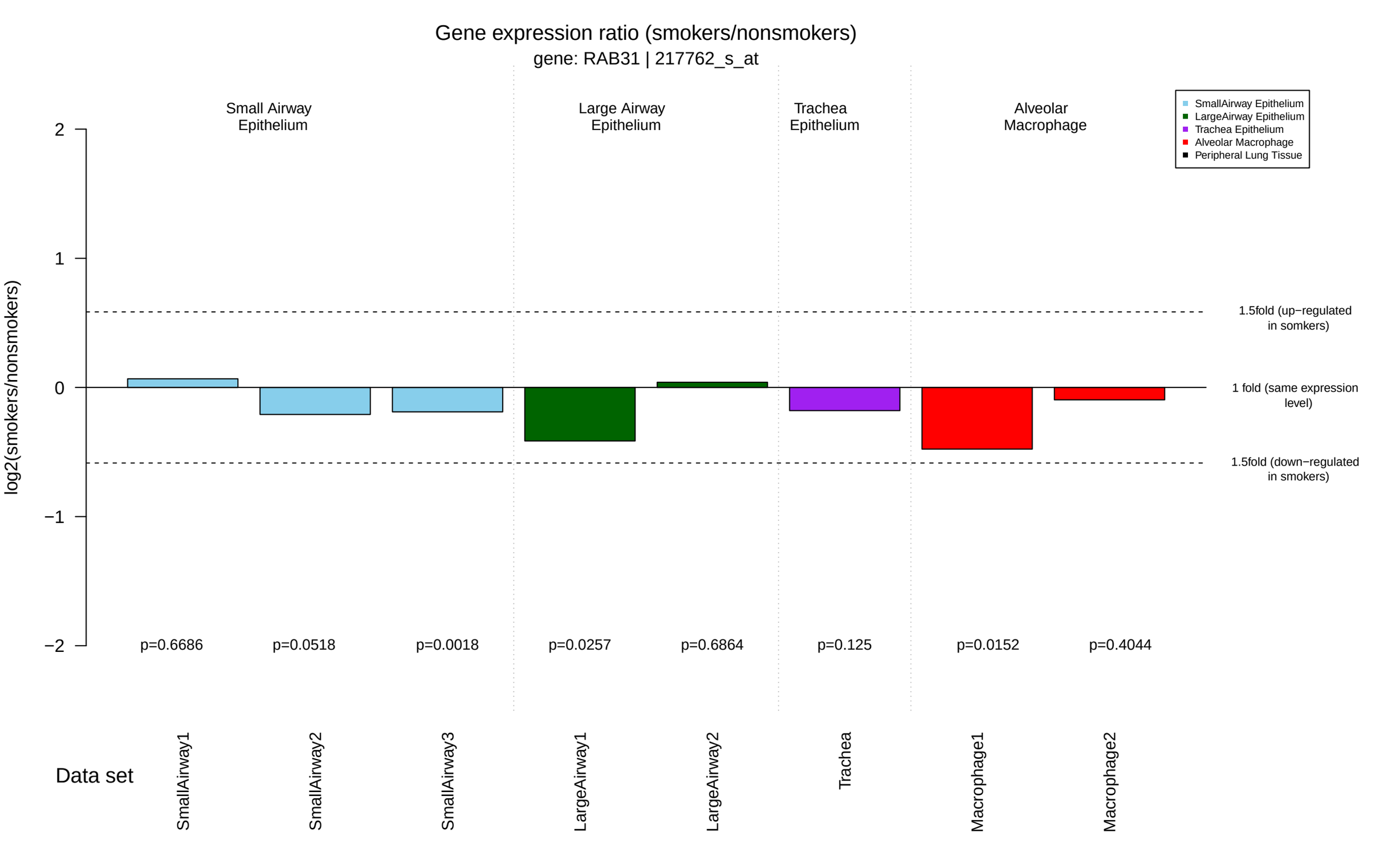
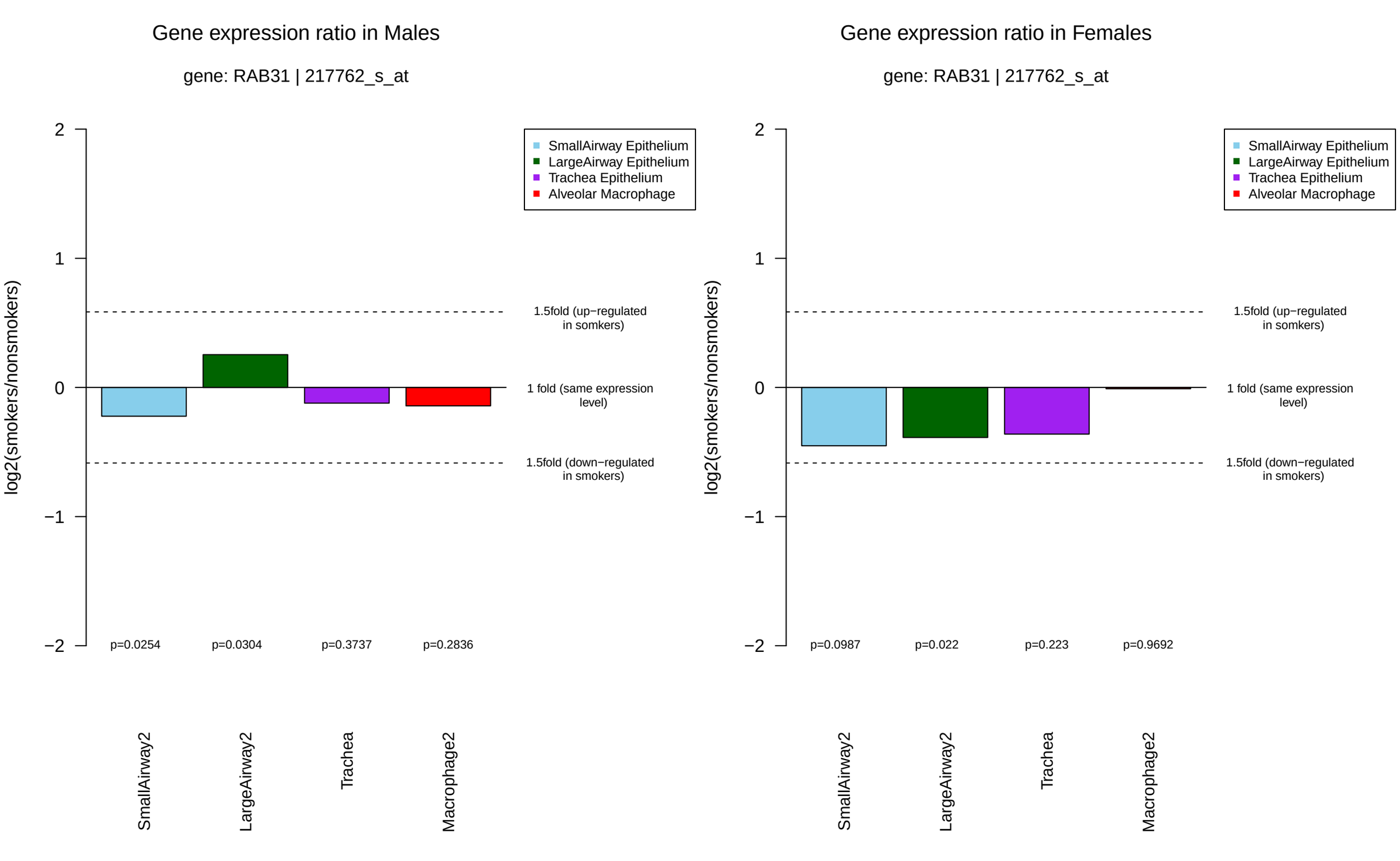
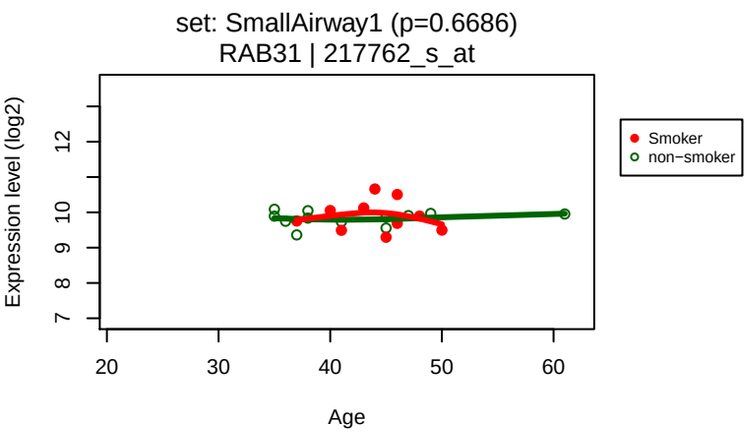
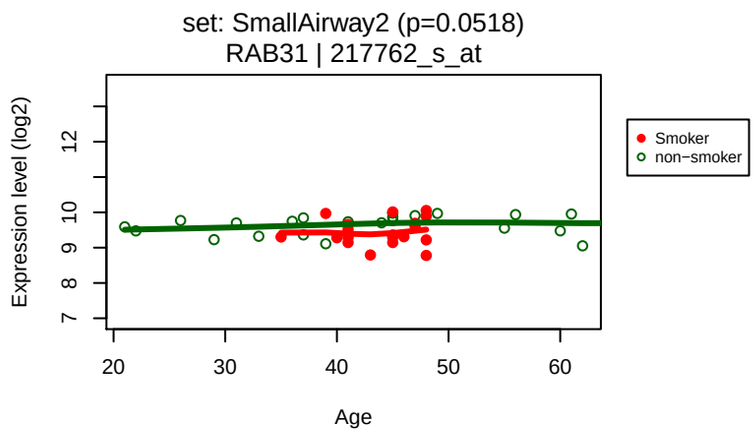
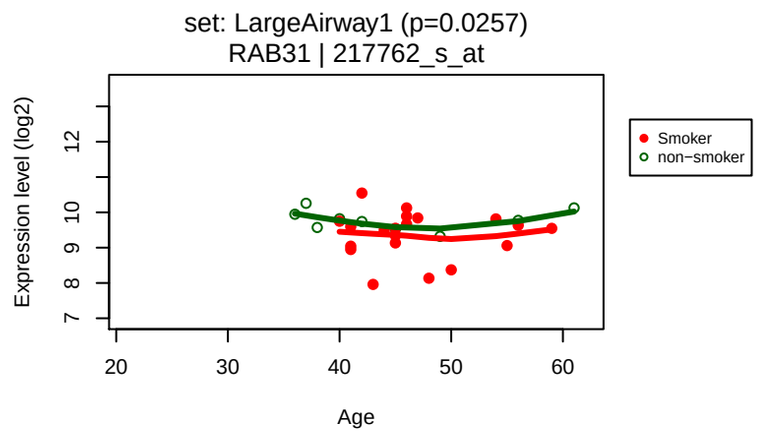

Gene expression of smokers and nonsmokers
gene: RAB31 | 217762_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.6686
p=0.0518
p=0.0018
p=0.0257
p=0.6864
p=0.125
p=0.0152
p=0.4044
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

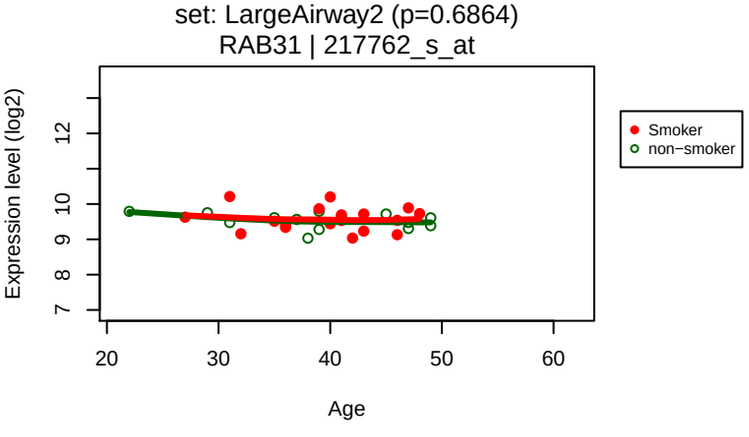

Gene expression ratio (smokers/nonsmokers)
gene: RAB31 | 217762_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6686
p=0.0518
p=0.0018
p=0.0257
p=0.6864
p=0.125
p=0.0152
p=0.4044
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

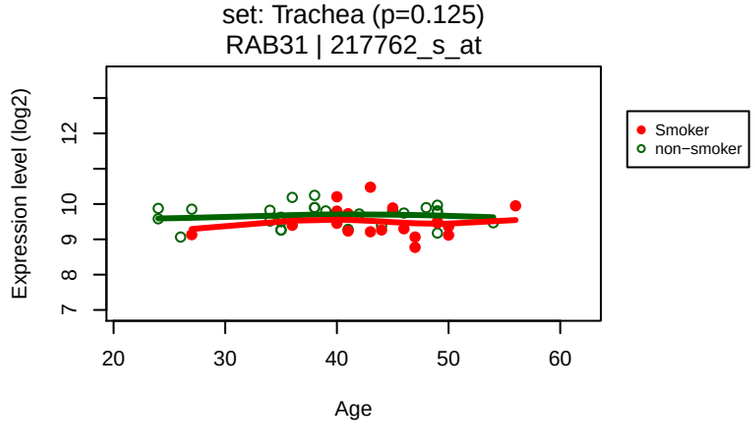

Gene expression ratio in Males
gene: RAB31 | 217762_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0254
p=0.0304
p=0.3737
p=0.2836
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RAB31 | 217762_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0987
p=0.022
p=0.223
p=0.9692
SmallAirway2
LargeAirway2
Trachea
Macrophage2

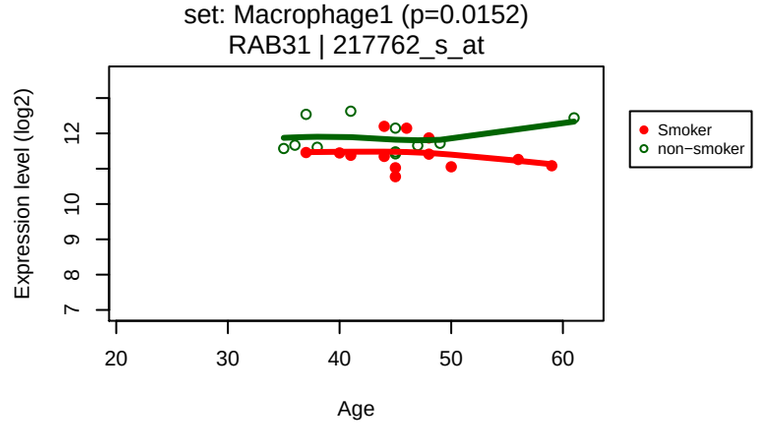

set: SmallAirway1 (p=0.6686)
RAB31 | 217762_s_at
Expression level (log2)
Age
Smoker
non−smoker

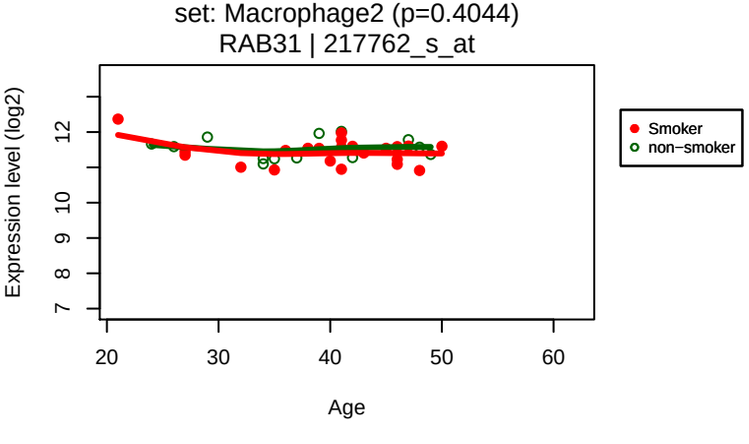

set: SmallAirway2 (p=0.0518)
RAB31 | 217762_s_at
Expression level (log2)
Age
Smoker
non−smoker

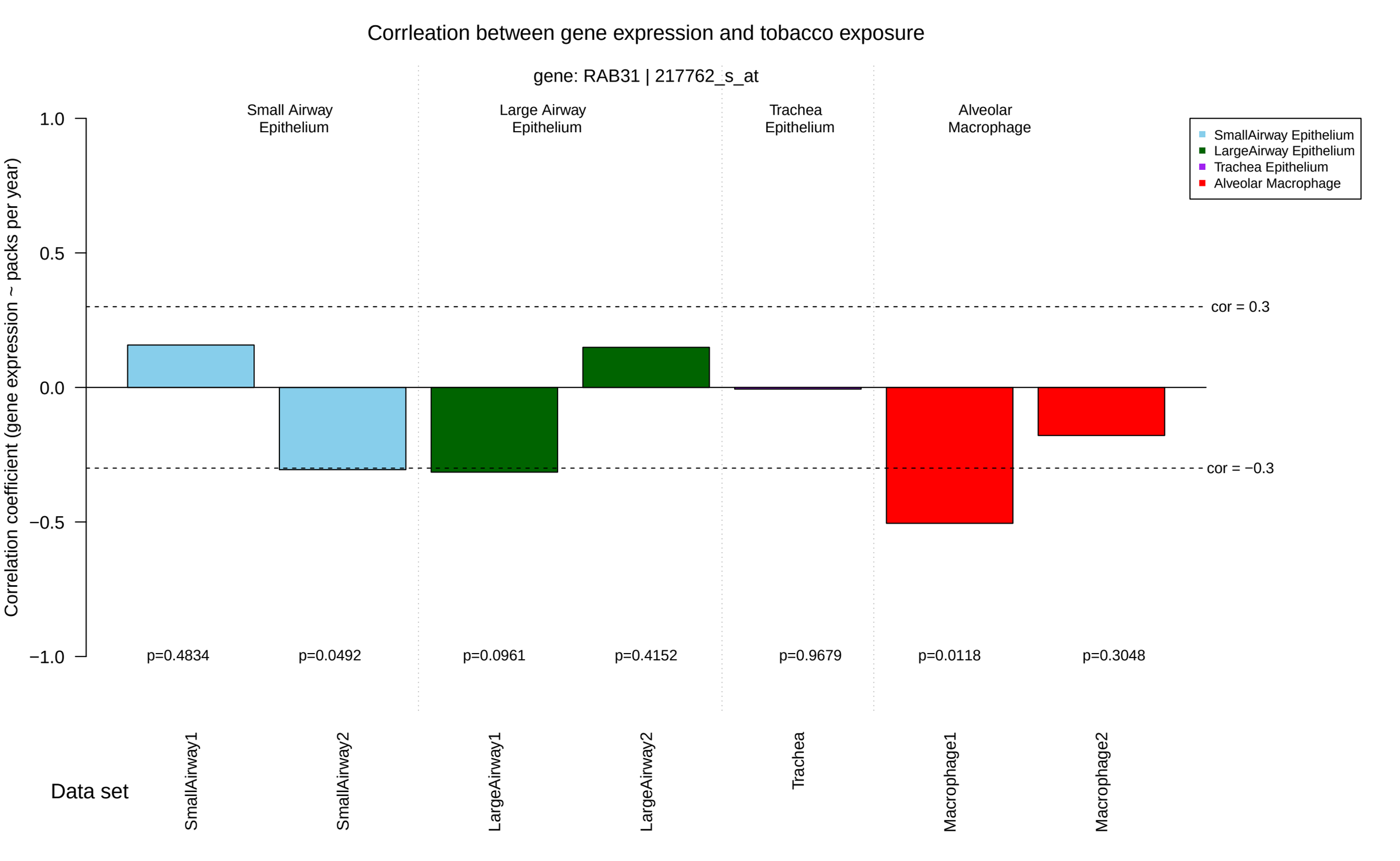

set: LargeAirway1 (p=0.0257)
RAB31 | 217762_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6864)
RAB31 | 217762_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.125)
RAB31 | 217762_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0152)
RAB31 | 217762_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4044)
RAB31 | 217762_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RAB31 | 217762_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
Correlation coefficient (gene expression ~ packs per year)
p=0.4834
p=0.0492
p=0.0961
p=0.4152
p=0.9679
p=0.0118
p=0.3048
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2